Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/29/2026
PI Name:	Estep, Patrick M.S.		
Project Title:	Characterization of Physical and Cognitive Workloads Associated with Frequent Exploration Surface Extravehicular Activities and Operations		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations (2) **Food and Nutrition**:Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition (3) **Sleep**:Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload (4) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	patrick.n.estep@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	NASA Johnson Space Center		
PI Address 1:	2101 E NASA Pwky		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	08/01/2024	**End Date:**	09/30/2027
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Co-Investigators (Co-Is) were updated through the reporting period. Steven Anderson left NASA and was replaced by Morgan Stosic. Dillon Frisco was replaced by Jonathan Hummel in support of exercise-focused measures and data analyses. Douglass Diak was added due to relevant expertise for immunology-focused measures.		
COI Name (Institution):	Bell, Suzanne Ph.D. (NASA Johnson Space Center) Crucian, Brian Ph.D. (NASA Johnson Space Center) Douglas, Grace Ph.D. (NASA Johnson Space Center) Landon, Lauren Ph.D. (KBR/NASA Johnson Space Center) Moudy, Sarah Ph.D. (MEI Technologies, Inc) Prejean, Brian Ph.D. (NASA Johnson Space Center) Rhodes, Richard M.S. (NASA Johnson Space Center) Schlotman, Taylor Ph.D. (NASA Johnson Space Center) Smith, Scott Ph.D. (NASA Johnson Space Center)		

	Young, Millenia Ph.D. (NASA Johnson Space Center) Zwart, Sara Ph.D. (NASA Johnson Space Center) Strock, Nicole Ph.D. (NASA Johnson Space Center) Marshall-Goebel, Karina Ph.D. (NASA Johnson Space Center) Stosic, Morgan (NASA Johnson Space Center) Hummel, Jonathan (NASA Johnson Space Center) Diak, Douglass (NASA Johnson Space Center)
Grant/Contract No.:	Directed Research
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Current nominal operations on the International Space Station (ISS) include roughly four microgravity extravehicular activities (EVAs) over the course of a 6-month mission. During the Space Shuttle Program, there were occasions on which astronaut crews conducted multiple EVAs in a week; however, those were unplanned short-duration contingency missions in a microgravity environment and associated spacesuit and were rarely performed on back-to-back days. Current lunar and planetary exploration mission designs, including Artemis missions, plan to increase the EVA operational tempo up to 24 hours of surface EVA per crewmember per week for consecutive weeks with unknown human health and performance implications. Further, aerobic capacity and muscle strength requirements to successfully complete high tempo surface EVA operations are not well understood or validated. This proposed study seeks to characterize physical and cognitive workloads, functional fitness requirements, and recovery needs associated with high-tempo exploration EVA operations. This study will be conducted in three primary phases. The first phase will assess and validate a simulated EVA task circuit as a predictor of EVA readiness and performance using simulated spacesuits, tools, and tasks in a 1-g setting. The second and third phases will assess and characterize impacts to human health and performance when completing a high operational tempo of multiple simulated EVAs in accordance with NASA's exploration concepts for operations; the second phase will utilize the same simulation concepts, tools, and methods as the first phase but conducted four times in five days, while the third phase will seek to leverage actual spacesuits at an elevated operating pressure and a high-fidelity partial gravity analog at the same tempo. This study will report integrated human health and performance-driven impacts and outcomes as a function of EVA duration and frequency, particularly up to 24 hours of simulated EVA in a week. Resulting measures will span multiple domains across operations, EVA, exercise, physiology, nutrition, immunology, cognitive performance, and neuroscience. We hypothesize that measurable decrements to human health and performance parameters will be observed in varying degrees across these domains but will not hinder ability to complete EVA objectives throughout the mission profile.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	Data collected during this study will help inform outcomes and considerations associated with ground-based EVA training for astronauts, and may also help supplement the existing knowledgebase for human health and performance impacts in high performing tactical athlete-type populations.
Task Progress:	In this reporting period (second year of grant), we have coordinated with the various stakeholders (NASA Human Research Program (HRP), Extravehicular Activity and Human Surface Mobility Program (EHP), NBL, and Co-Investigators) to initiate the study in September 2025. [Note: The investigation includes three study groups: the HS3 Singles group, the HS3 Hi-Tempo group, and the NBL Hi-Tempo group (Ed., 6/5/26).] Recruitment and data collection started for the HS3 singles arm of the study, in which 6 subjects have completed all testing (as of 14 May 2026). Work for the NBL Hi-Tempo group was expedited, and enrollment has proceeded, with 5 consented crew subjects. Several modifications to the Institutional Review Board (IRB) protocol have been submitted and approved. The initial data analysis and data management plans were drafted. Operator training and certification to perform HS3 singles testing was completed. There is ongoing work to develop and finalize the test plan, developmental engineering runs, and test readiness review for the initiation of the NBL Hi-Tempo data collection to begin in July 2026.
Bibliography Type:	Description: (Last Updated: 06/05/2026)
Abstracts for Journals and Proceedings	Estep P, Strock N. "Hi-Tempo EVA Study: Characterization of physical and cognitive workloads associated with frequent exploration surface extravehicular activities. and operations." " 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026